HACKERVILLA NAINITAL

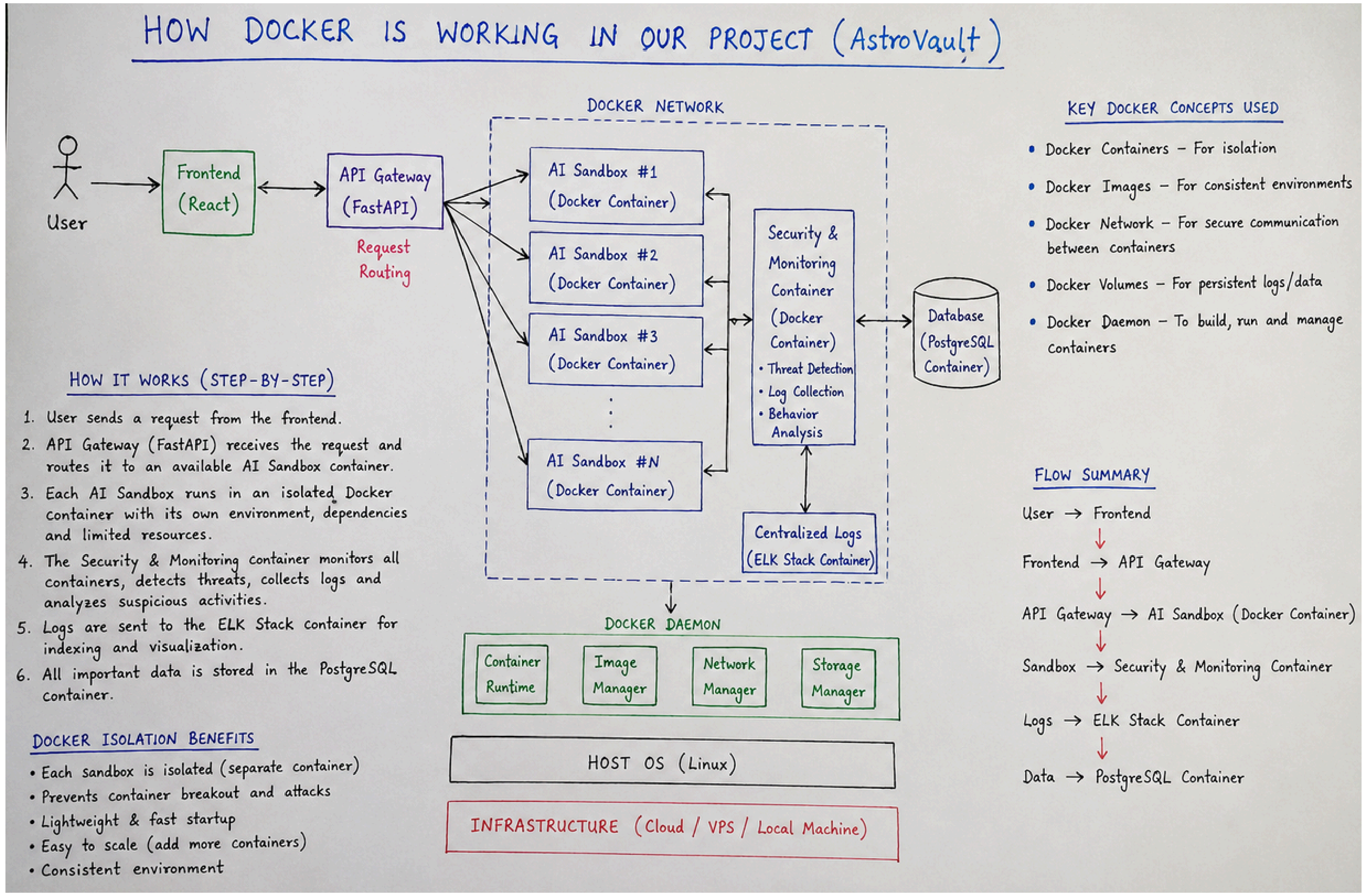
- **Team Name - Reputation & Graduation**
- **Members’ Name - Arshal , kanak , Utkarsh , Ankit**
- **DEmo link :- https://www.youtube.com/watch?v=LncHWck3yGA**

Problem Statement

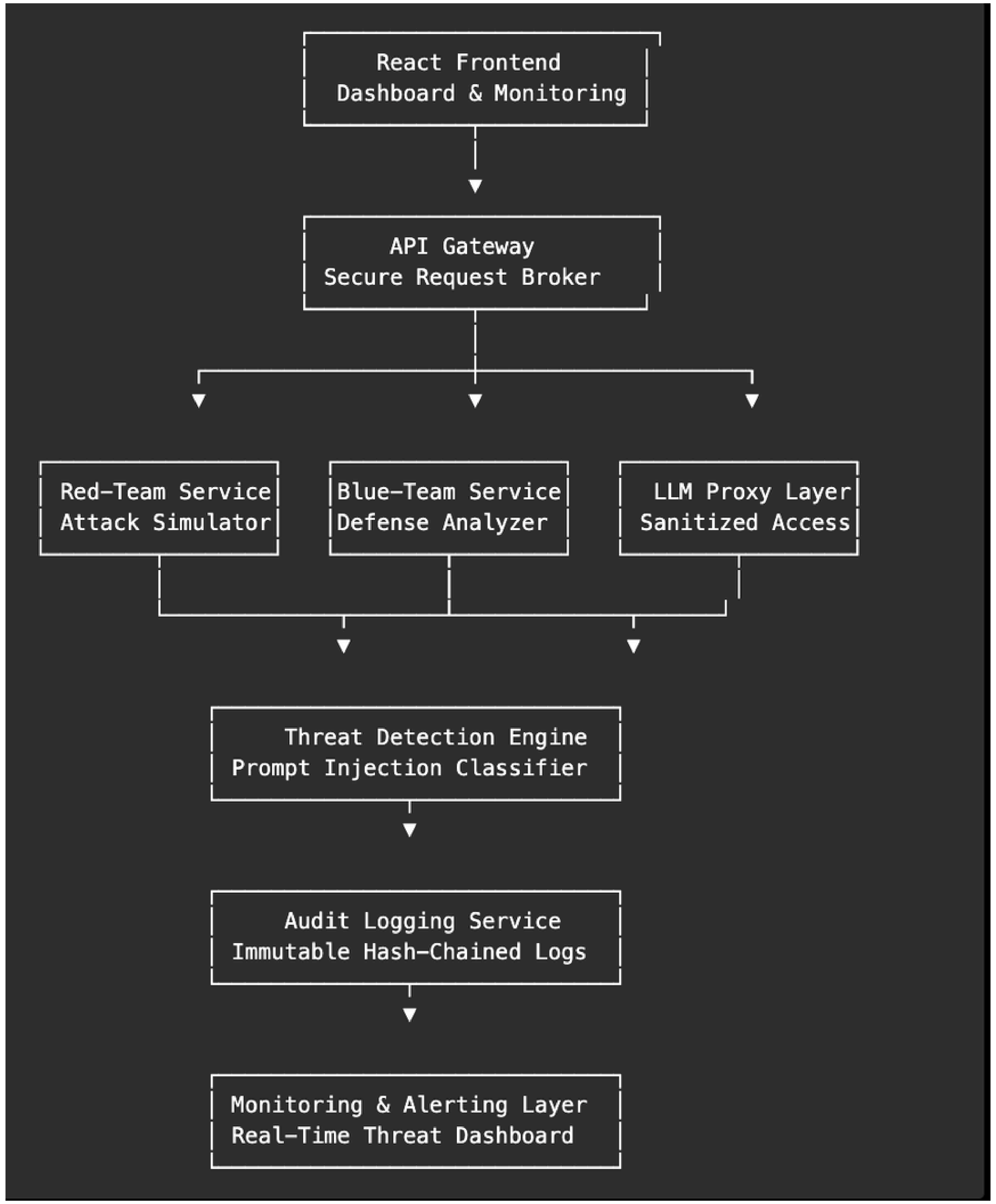
- **Modern AI systems are increasingly being deployed in high-risk environments such as finance, healthcare, public infrastructure, and enterprise automation. To ensure these systen are safe, organizations conduct adversarial AI testing using red-team and blue-team workflows.**
- **IMPACT :-**
- **1. Trustworthy AI Safety Evaluation**
- **2. Secure Multi-Tenant AI Testing**
- **3. Prevention of Prompt Leakage & Jailbreak Contamination**
- **4. Real-Time Threat Detection**
- **5. Tamper-Evident Audit Integrity**

Solution

- Solution Overview :-Proposed Solution
- AstroVault is a secure adversarial AI testing platform built using a multi-layer isolation architecture.
- Core Components
- 1. Secure API Gateway
- Acts as the central broker that validates requests, manages sessions, enforces access control, and routes traffic securely.
- 2. Isolated Team Containers
- Separate Docker containers are used for:
- Red-team attack simulations
- Blue-team defensive analysis
- LLM proxy execution
- Each container operates with restricted permissions and isolated communication boundaries.
- 3. Threat Detection Engine
- An AI-assisted detection layer classifies:
- jailbreak attempts,
- prompt injections,
- suspicious instructions,
- and data exfiltration attempts.
- 4. LLM Proxy Layer
- Prevents direct model access and sanitizes prompts before inference execution.
- 5. Immutable Audit Logging
- Every interaction is logged with timestamping and cryptographic hash chaining for tamper detection.
- 6. Real-Time Monitoring Dashboard
- Provides live visibility into:
- blocked attacks,
- risk scores,
- isolation violations,
- and suspicious activity.



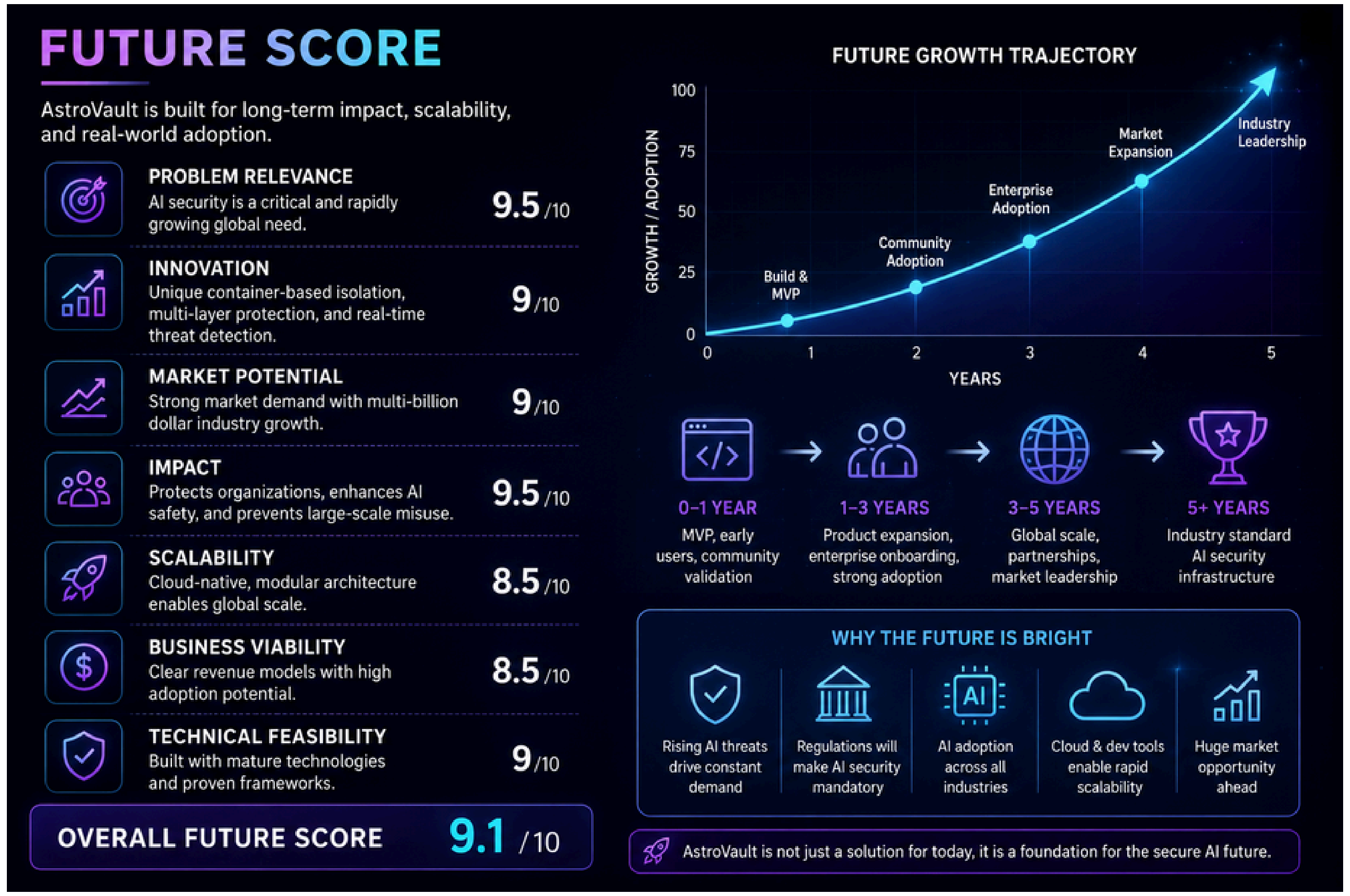
Architecture



- **Tech stack :- next js , node , docker for environment , clerk for auth , jwt for bypassing , sha256 for scripting**
- **APIs used : -normal for GEMINI 2.5 flash for redirecting and traiing the thing**

Future scope and where we are leading to

- Target users => AI chatbots , AI copilots , autonomous AI agents , LLM-based SaaS tools , AI Safety & Research Organizations , Cybersecurity Teams
- Market size :- SOM Generative AI Cybersecurity Market (~21%–34% CAGR)
- Potential revenue model : aaS Subscription Model (Primary Model) , API-as-a-Service , Enterprise Self-Hosted Deployment , AI Security Auditing Services , Educational & Research Licensing
- Expected growth :- apid enterprise AI adoption,
- increasing AI-powered cyber threats,
- secure deployment requirements,
- and demand for trustworthy AI systems.



Future scope

- Mobile app deployment :- AstroVault can later integrate with enterprise cybersecurity and cloud infrastructure tools such as SIEM platforms, authentication systems, monitoring dashboards, and internal AI governance pipelines. This would make the platform easier to adopt in large-scale enterprise environments. (very less chances)
- AI model improvement :- The platform can integrate more advanced AI detection models trained specifically for prompt injection, jailbreak attempts, adversarial prompts, and suspicious AI interactions. Continuous improvements in the threat detection engine will increase classification accuracy and reduce false positives over time.
- Enterprise integrations :-AstroVault can later integrate with enterprise cybersecurity and cloud infrastructure tools such as SIEM platforms, authentication systems, monitoring dashboards, and internal AI governance pipelines. This would make the platform easier to adopt in large-scale enterprise environments.
- Scaling to global users : Since AstroVault is built using a cloud-native and containerized architecture, the platform can be scaled across multiple cloud regions and providers. This would enable support for global organizations running large-scale AI systems and multi-tenant adversarial testing environments.